Project

1. Prepare a frequency table of the top speeds of 20 different land animals. Find mean, median and mode. Justify your answer.

2. From the record of students particulars of the class,

   (i) Find the mean age of the class (using class interval)

   (ii) Calculate the mean height of the class(using class intervals)

Points to Remember

- The information collected for a definite purpose is called data.
- The data collected by the investigator are known as primary data. When the information is gathered from an external source,the data are called secondary data.
- Initial data obtained through unorganized form are called Raw data.
- Mid Value $= \frac{UCL + LCL}{2}$ (where UCL–Upper Class Limit, LCL–Lower Class Limit).
- Size of the class interval $= UCL - LCL$.
- The mean for grouped data:

Direct Method	Assumed Mean Method	Step-Deviation Method
$\bar{X} = \frac{\sum fx}{\sum f}$	$\bar{X} = A + \frac{\sum fd}{\sum f}$	$\bar{X} = A + \left[\frac{\sum fd}{\sum f} \times c\right]$

- The cumulative frequency of a class is the frequency obtained by adding the frequency of all up to the classes preceeding the given class.
- Formula to find the median for grouped data: Median $= l + \frac{\left(\frac{N}{2} - m\right)}{f} \times c$.
- Formula to find the mode for grouped data: Mode $= l + \left(\frac{f - f_1}{2f - f_1 - f_2}\right) \times c$.

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